

Register No: 192421231

Student: YUVASRI B

Assigned Question:

Q40. Debugging Insertion Sort Code (Incorrect Final Placement Index)

The following Insertion Sort implementation performs element shifting, but the final insertion step uses the wrong index and causes misplaced values in the sorted prefix.

Carefully analyze the shifting process and identify the exact source of the indexing error.

Debug the code step-by-step so that each element is inserted into its correct position after all larger elements are shifted. Optimize the implementation for readability and maintain consistent variable usage. Analyze the corrected algorithm in terms of time complexity and rewrite the final clean version with proper comments.

```
def insertion_sort(arr):  
    for i in range(1, len(arr)):  
        key = arr[i]  
        j = i - 1  
        while j >= 0 and arr[j] > key:  
            arr[j+1] = arr[j]  
            j -= 1  
        arr[j] = key # ERROR  
    return arr
```

Output:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aee6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/DAA/insertion.py
Sorted Array: [5, 5, 5, 5, 1]
>>>
```

Error:

Wrong insertion index.

Correct:

`arr[j+1] = key`

Debug the Program:

Replace

`arr[j] = key`

with

`arr[j+1] = key`

Corrected output:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
- RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/DAA/insertion.py
Sorted Array: [5, 5, 5, 5, 1]
>>>
===== RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/DAA/insertion.py =====
Original Array: [5, 3, 4, 1, 2]
Sorted Array: [1, 2, 3, 4, 5]
>>>
```

Error Identified:

- The insertion statement used `arr[j] = key`, which inserts the key at the wrong index.

Correction:

- Replaced `arr[j] = key` with `arr[j + 1] = key`.

Optimization:

- Added comments for better readability.
- Maintained proper indentation.
- Used consistent variable names (key, j, arr).
- The algorithm remains efficient and easy to understand.

Time Complexity:

- Best Case: **$O(n)$**
- Average Case: **$O(n^2)$**
- Worst Case: **$O(n^2)$**
- Space Complexity: **$O(1)$**